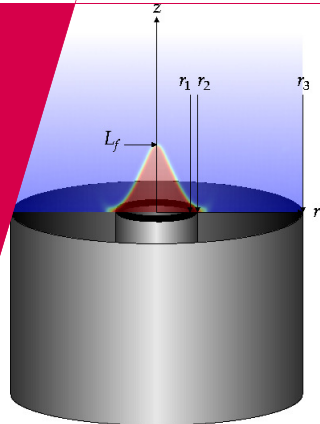


Givens rotations

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Givens rotations

Definition and properties of Givens rotations

Choosing Givens rotations

Summary

Introduction to Givens rotations

Householder transformation:

- ▶ Creates zero elements at all positions below the diagonal in a column
- ▶ $n - 1$ transformations needed to obtain an upper triangular matrix

Givens rotation:

- ▶ Creates a zero element at only one position in a matrix
- ▶ Rotation needed for every nonzero element below the diagonal

$$\underbrace{\begin{pmatrix} * & * & * \\ * & * & * \\ * & * & * \end{pmatrix}}_{\mathbf{A}} \xrightarrow{\mathbf{G}_1} \underbrace{\begin{pmatrix} * & * & * \\ 0 & * & * \\ * & * & * \end{pmatrix}}_{\mathbf{G}_1\mathbf{A}} \xrightarrow{\mathbf{G}_2} \underbrace{\begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}}_{\mathbf{G}_2\mathbf{G}_1\mathbf{A}} \xrightarrow{\mathbf{G}_3} \underbrace{\begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}}_{\mathbf{G}_3\mathbf{G}_2\mathbf{G}_1\mathbf{A}}$$

Define $\mathbf{R} := \mathbf{G}_3\mathbf{G}_2\mathbf{G}_1\mathbf{A}$. Then $\mathbf{R}$ is upper triangular and $\mathbf{A} = \mathbf{G}_1^T\mathbf{G}_2^T\mathbf{G}_3^T\mathbf{R}$

Define $\mathbf{Q} := \mathbf{G}_1^T\mathbf{G}_2^T\mathbf{G}_3^T$. Then $\mathbf{Q}$ is orthogonal and $\mathbf{A} = \mathbf{QR}$

Givens rotation

Definition (Givens rotation)

A *Givens rotation* is a matrix of the form

$$\mathbf{G} = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & c & & s & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 & \\ & & -s & & & c & & \\ & & & & & & 1 & \\ & & & & & & & \ddots & \\ & & & & & & & & 1 \end{pmatrix}$$

← row p

← row q

↑ column p ↑ column q

with $c^2 + s^2 = 1$

$\mathbf{G}$ is the identity matrix $\mathbf{I}$ except for rows and columns p and q

$$\begin{aligned} g_{pp} &= c & g_{pq} &= s \\ g_{qp} &= -s & g_{qq} &= c \end{aligned}$$

$\mathbf{G}$ is a rotation in the (p, q) -plane over an angle ϕ with

$$\begin{aligned} c &= \cos \phi \\ s &= \sin \phi \end{aligned}$$

$\mathbf{G}$ is orthogonal, since $\mathbf{G}^T \mathbf{G} = \mathbf{I}$

Choosing Givens rotations

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$

Write $\mathbf{A} = (\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n)$

We choose $\mathbf{G}_1$ such that

$$\mathbf{u} = \mathbf{a}_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ a_{n1} \end{pmatrix} \xrightarrow{\mathbf{G}_1} \mathbf{v} = \begin{pmatrix} \alpha \\ 0 \\ a_{31} \\ \vdots \\ a_{n1} \end{pmatrix}$$

Since $\mathbf{u}$ and $\mathbf{v}$ must have the same length:

$$\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 \implies \alpha = \sqrt{a_{11}^2 + a_{21}^2}$$

We use the Givens rotation

$$\mathbf{G}_1 = \begin{pmatrix} c & s & & & \\ -s & c & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}$$

The requirement $\mathbf{G}_1 \mathbf{u} = \mathbf{v}$ gives the equations:

$$ca_{11} + sa_{21} = \alpha$$

$$-sa_{11} + ca_{21} = 0$$

Multiply first equation with a_{11} ,
second with a_{21} and sum

$$ca_{11}^2 + sa_{11}a_{21} = \alpha a_{11}$$

$$-sa_{11}a_{21} + ca_{21}^2 = 0$$

$$\hline c(a_{11}^2 + a_{21}^2) = \alpha a_{11}$$

Since $a_{11}^2 + a_{21}^2 = \alpha^2$, we find

$$c = a_{11}/\alpha$$

Substitute in $-sa_{11} + ca_{21} = 0$

to get $s = a_{21}/\alpha$

Choosing Givens rotations (cont.)

$$\mathbf{a}_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \\ \vdots \\ a_{n1} \end{pmatrix} \xrightarrow{\mathbf{G}_1} \begin{pmatrix} \sqrt{a_{11}^2 + a_{21}^2} \\ 0 \\ a_{31} \\ a_{41} \\ \vdots \\ a_{n1} \end{pmatrix} \xrightarrow{\mathbf{G}_2} \begin{pmatrix} \sqrt{a_{11}^2 + a_{21}^2 + a_{31}^2} \\ 0 \\ 0 \\ a_{41} \\ \vdots \\ a_{n1} \end{pmatrix}$$

To transform column 1 to upper triangular form, we need $n - 1$ Givens rotations

2	$n - 2$
$\vdots$	$\vdots$
$n - 1$	1

Total number of Givens rotations needed:

$$(n - 1) + (n - 2) + \cdots + 3 + 2 + 1 = \sum_{k=1}^{n-1} k = \frac{1}{2}n(n - 1)$$

Example

Let $\mathbf{A} = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 1 & 0 \\ 3 & 2 & 3 \end{pmatrix}$

Find a QR decomposition for $\mathbf{A}$ using Givens rotations



Solution

Summary

In this video we have seen how you can

- ▶ Find a Givens rotation to create one zero in a matrix
- ▶ Compute the QR decomposition of a matrix using Givens rotations

